HIGHER DIRECT IMAGES R^N A_* UNDER AN AUGMENTATION AS SHEAVES OF COHOMOLOGY ON THE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the author opens with 'This is the analogue of Cohomology, Lemma ... and we strongly encourage the reader to skip the proof' (source lines 680-683) - the proof is nonetheless given in full and nothing is omitted; the exactness of the extension-by-zero functor j_! is invoked as a standard property (source line 706), matching the hypothesis of the cited Homology lemma; the equalizer description of a_* used in (2a) is established on card L46175 (lemma-augmentation).

Editorial note. The label lemma-augmentation-pushforward-higher-direct-image names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The canonical topos morphism attached to an augmentation. Same chapter, \label{lemma-augmentation} (card L46175). Proof lines 694-696 (2a): a_*F is the equalizer of the two maps a_{0,*}F_0 -> a_{1,*}F_1.
- Cohomology of sheaves on the total simplicial site. Same chapter, \label{lemma-simplicial-sheaf-cohomology}. Proof lines 701-704 (2d): H^0 of the total-site section (X x_Y V)_Zar is the equalizer of H^0(X_0 x_Y V, F_0) -> H^0(X_1 x_Y V, F_1).
- The model description of higher direct images. Cohomology, Lemma \label{cohomology-lemma-describe-higher-direct-images} and Cohomology on Sites, Lemma \label{sites-cohomology-lemma-higher-direct-images}. Proof lines 680-683: the lemma is the analogue of these descriptions of the higher direct images; the author frames it as such and gives the proof in full.
- Injectivity is preserved by exact left adjoints. Homology, Lemma \label{homology-lemma-adjoint-preserve-injectives}. Proof lines 706-710 (1): the extension-by-zero functor j_! : Ab((X x_Y V)_Zar) -> Ab(X_Zar), exact left adjoint to restriction, preserves injectives.

Source reference. Original lemma 01E4: sources/stacks-chow/tags.txt, source lines 1821–1821. Prerequisite pointer invoked at proof line(s) 681; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 015Z: sources/stacks-chow/tags.txt, source lines 1526–1526. Prerequisite pointer invoked at proof line(s) 710; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09W4: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 524–524. Prerequisite pointer invoked at proof line(s) 696; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09W6: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 624–624. Prerequisite pointer invoked at proof line(s) 704; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 072W: sources/stacks-chow/tags.txt, source lines 9040–9040. Prerequisite pointer invoked at proof line(s) 682; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a simplicial space and let $a : X \to Y$ be an augmentation. Let $\mathcal{F}$ be an abelian sheaf on X_{Zar}. Then $R^n a_* \mathcal{F}$ is the sheaf associated to the presheaf*

$$V \longmapsto H^n((X \times_Y V)_{Zar}, \mathcal{F}|_{(X \times_Y V)_{Zar}})$$

Proof. This is the analogue of Cohomology, Lemma 01E4 or of Cohomology on Sites, Lemma 072W and we strongly encourage the reader to skip the proof. Choosing an injective resolution of $\mathcal{F}$ on X_{Zar} and using the definitions we see that it suffices to show: (1) the restriction of an injective abelian sheaf on X_{Zar} to $(X \times_Y V)_{Zar}$ is an injective abelian sheaf and (2) $a_* \mathcal{F}$ is equal to the rule

$$V \longmapsto H^0((X \times_Y V)_{Zar}, \mathcal{F}|_{(X \times_Y V)_{Zar}})$$

Part (2) follows from the following facts

(2a) $a_* \mathcal{F}$ is the equalizer of the two maps $a_{0,*}\mathcal{F}_0 \to a_{1,*}\mathcal{F}_1$ by Lemma 09W4,
(2b) $a_{0,*}\mathcal{F}_0(V) = H^0(a_0^{-1}(V), \mathcal{F}_0)$ and $a_{1,*}\mathcal{F}_1(V) = H^0(a_1^{-1}(V), \mathcal{F}_1)$,
(2c) $X_0 \times_Y V = a_0^{-1}(V)$ and $X_1 \times_Y V = a_1^{-1}(V)$,
(2d) $H^0((X \times_Y V)_{Zar}, \mathcal{F}|_{(X \times_Y V)_{Zar}})$ is the equalizer of the two maps $H^0(X_0 \times_Y V, \mathcal{F}_0) \to H^0(X_1 \times_Y V, \mathcal{F}_1)$ for example by Lemma 09W6.

Part (1) follows after one defines an exact left adjoint $j_! : Ab((X \times_Y V)_{Zar}) \to Ab(X_{Zar})$ (extension by zero) to restriction $Ab(X_{Zar}) \to Ab((X \times_Y V)_{Zar})$ and using Homology, Lemma 015Z. □